Check-In #2

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Structural Comparison of Quantum Ansatz for Ground-State Approximation via Subspace Projection

Summary of progress. Since Check-In #1, I completed a fair subspace-benchmark workflow for three ansatz families (Krylov, UCCSD, fixed HEA), together with exact FCI reference energies and a first PennyLane-based resource-estimation study. Relative to the original proposal, I replaced H_2O with LiH as the largest validated benchmark because the current $H_2/H_4/LiH$ set can be run end-to-end with exact cross-checks and reproducible figures.

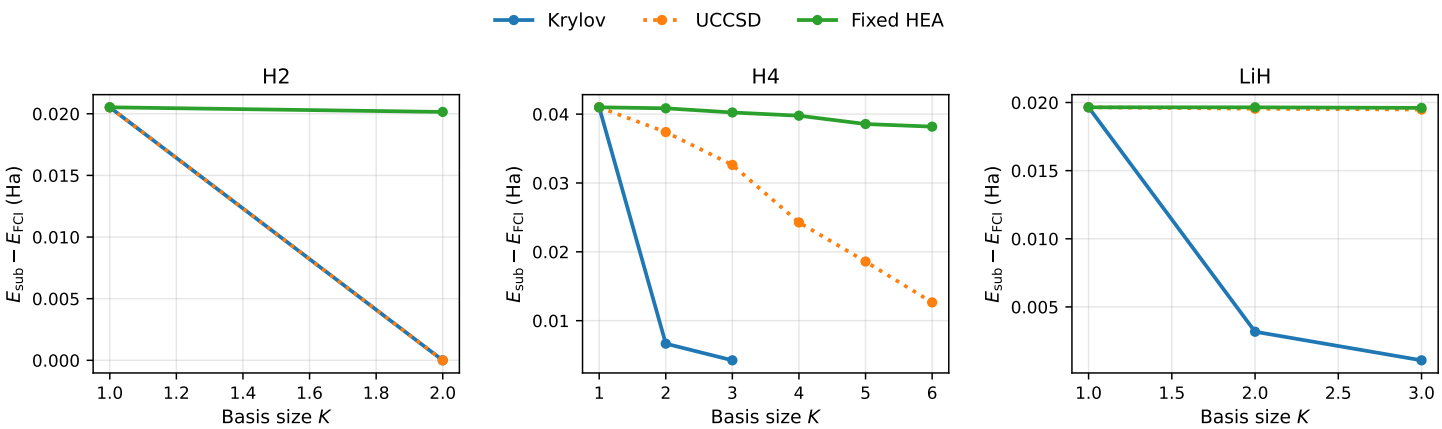
Implemented now

- Exact FCI reference workflow for H_2, H_4, and LiH.
- Separate Krylov, UCCSD, and fixed-HEA basis-state generation notebooks.
- Fair subspace benchmark notebooks that compare the three ansatz families at fixed basis size K.
- Preliminary PennyLane resource-estimation workflow using both `qml.specs(...)` and `pennylane.estimator.estimate(...)`.

Benchmark systems

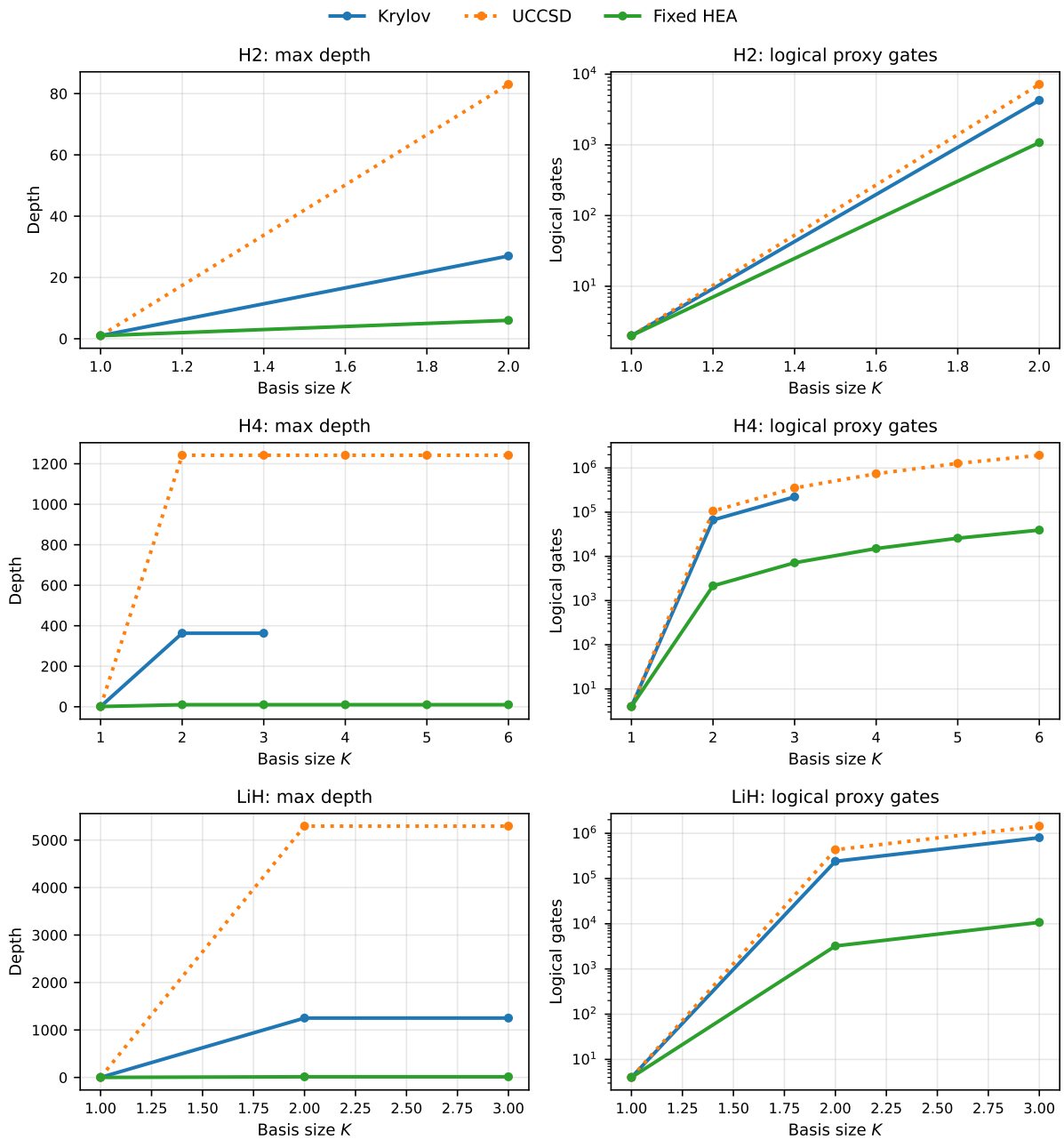
System	Bond (Å)	Qubits	Electrons	Pauli terms
H_2	0.74	4	2	15
H_4	0.74	8	4	185
LiH	1.5474	12	4	631

Preliminary results. The current gap-vs-K trends already separate the ansatz families. Krylov is the strongest low-K basis in all three systems, UCCSD shows the clearest continued improvement for H_4, and LiH remains a harder optimization problem for the parameterized ansatz families under the current budget.



Updated plan toward the final report. My original proposal and Check-In #1 emphasized implementing Hadamard-test matrix-element estimation. After building the ansatz-state workflows, I think that is not the most informative comparison axis for this project. Once a basis state is prepared, the downstream overlap/Hamiltonian-estimation workflow is largely shared across Krylov, UCCSD, and HEA, so a Hadamard-test implementation would not isolate the main ansatz-dependent difference very well. The more meaningful comparison is therefore *basis quality plus state-preparation cost*. The updated path is: finalize the fair subspace benchmarks, use the energy-gap trends as preliminary expressivity evidence, and use PennyLane-based resource estimation for the stack-level implications.

Resource estimation. I now have a combined notebook, `Resource_Estimation_Subspace.ipynb`, that evaluates the selected basis states from the validated subspace benchmarks without rerunning the expensive optimization. It uses two PennyLane resource layers: `qml.specs(...)` for native circuit metrics such as gate count and depth, and `pennylane.estimator.estimate(...)` for logical gate-set estimates. The resource study uses frozen selected basis states and a prep-only subspace proxy, so the current results already connect ansatz choice to circuit depth and total subspace-evaluation cost.



Draft Full-Stack Impact / Resource Implications. The preliminary full-stack conclusion is that the ansatz choice changes the cost mainly through state preparation, not just through the generalized eigensolver that comes afterward. Krylov uses the smallest basis-state parameterization, but its Trotterized evolution can still accumulate a large logical cost as the system grows. UCCSD becomes the heaviest native and logical option for the larger systems, especially LiH, while the current fixed HEA remains much shallower and lighter under its one-layer architecture. At the same time, the total subspace cost grows quickly with basis size because the number of unique overlap and Hamiltonian matrix elements scales quadratically in K: $\#S_{\text{offdiag}} = K(K-1)/2$ and $\#H_{\text{unique}} = K(K+1)/2$. This means the final report should compare ansatz families not only by achieved energy gap but also by the resource implications of reaching that subspace quality.

Prototype and validation status. The current notebooks already provide an end-to-end prototype on meaningful examples and preliminary validation against exact FCI references, which gives a concrete base for the draft report and the final reproducibility package.

Repository links. GitHub repository: github.com/jhp1859/p759_final_project. Main notebook links: H_2 fair comparison, H_4 fair comparison, LiH fair comparison, and resource estimation.